

Tutorial 10

Exercise 1 – Bayesian inference with a conjugate prior

Consider an i.i.d. dataset

$$X = \{x_i\}_{i=1,\dots,N}, \quad x_i \in \mathbb{R}, \quad \text{such that} \quad x_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2),$$

where σ^2 is assumed known. We place a Gaussian prior on μ :

$$\mu \sim \mathcal{N}(\mu_0, \lambda^2).$$

The goal of this exercise is to understand the limiting behavior of the posterior and predictive distributions as a function of the prior and the data.

We denote

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \tau = \frac{1}{\sigma^2}, \quad \tau_0 = \frac{1}{\lambda^2}.$$

In this model, the prior is conjugate to the likelihood and therefore admits an analytical expression :

$$p(\mu \mid X) = \mathcal{N}(\mu_N, \tau_N^{-1}), \quad \text{with} \quad \tau_N = \tau_0 + N\tau, \quad \mu_N = \frac{\tau_0\mu_0 + N\tau\bar{x}}{\tau_0 + N\tau}.$$

We also admit that the predictive distribution for a new observation $\tilde{x}$ is

$$p(\tilde{x} \mid X) = \mathcal{N}(\mu_N, \sigma^2 + \tau_N^{-1}).$$

Q 1.1 Write Bayes' rule for $p(\mu \mid X)$. What is, in general, the main difficulty in computing this quantity ?

Q 1.2 Show that the posterior mean μ_N can be written as a convex combination of μ_0 and $\bar{x}$. Interpret the relative role of the prior and the data.

Q 1.3 Give the parameter μ_{MAP} maximizing $p(\mu \mid X)$. Compare μ_{MAP} to the posterior mean. Can the predictive distribution be estimated with MAP, and what are its limitations ?

Q 1.4 Interpret the terms in the predictive variance. Also study the following limiting behaviors of μ_N and the posterior variance τ_N^{-1} :

- $N \rightarrow \infty$,
- $\lambda^2 \rightarrow \infty$ (weakly informative prior),
- $\sigma^2 \rightarrow \infty$.

Exercise 2 – Kullback–Leibler divergence

Let p and q be two probability distributions defined on a space $\mathcal{X}$.

The Kullback–Leibler divergence is defined by

$$\text{KL}(q\|p) = \mathbb{E}_q \left[\log \frac{q(x)}{p(x)} \right] = \int q(x) \log \frac{q(x)}{p(x)} dx.$$

Q 2.1 Show that

$$\text{KL}(q\|p) \geq 0.$$

Hint : use Jensen's inequality for a concave function φ :

$$\mathbb{E}[\varphi(X)] \leq \varphi(\mathbb{E}[X]).$$

Q 2.2 Show that

$$\text{KL}(q\|p) = 0 \iff q = p \text{ (presque partout)}.$$

Q 2.3 Show that the KL divergence is not symmetric, that is, in general,

$$\text{KL}(q\|p) \neq \text{KL}(p\|q).$$

Q 2.4 Consider now a multimodal distribution p , for instance a mixture of two well-separated Gaussians. We want to find an approximation q of p by minimizing the KL divergence. Compare qualitatively the solutions obtained by minimizing $\text{KL}(q\|p)$ and $\text{KL}(p\|q)$. Draw a figure illustrating the two behaviors. Give a practical reason why one usually chooses to optimize $\text{KL}(q\|p)$ rather than $\text{KL}(p\|q)$ in machine learning.

Exercise 3 – Mean-field variational inference

Consider the following model

$$p(x, z) = p(x|z)p(z),$$

where $z = (z_1, z_2)$ is a latent variable.

We want to approximate the posterior distribution $p(z|x)$ by a factorized distribution :

$$q(z) = q_1(z_1) q_2(z_2).$$

We define the ELBO :

$$\mathcal{L}(q) = \mathbb{E}_q[\log p(x, z)] - \mathbb{E}_q[\log q(z)].$$

Q 3.1 Show that maximizing the ELBO is equivalent to minimizing

$$\text{KL}(q(z)\|p(z|x)).$$

Q 3.2 Assuming q_2 is fixed, show that the optimum with respect to q_1 satisfies :

$$\log q_1(z_1) = \mathbb{E}_{q_2}[\log p(x, z)] + \text{const.}$$

Q 3.2.1 Show that maximizing the ELBO can be written as an optimization problem of the form

$$\max_{q_1} \int q_1(z_1) f(z_1) dz_1 - \int q_1(z_1) \log q_1(z_1) dz_1$$

under the constraint

$$\int q_1(z_1) dz_1 = 1.$$

Q 3.2.2 Write the Lagrangian $J(q_1, \lambda)$ associated with this optimization problem

Q 3.2.3 Assume that the functional derivative of J with respect to q_1 is given by

$$\frac{\delta J}{\delta q_1(z_1)} = f(z_1) - (\log q_1(z_1) + 1) + \lambda.$$

Deduce the relation satisfied by $\log q_1(z_1)$ at the optimum.

Q 3.3 Give the analogous expression for q_2 .

Q 3.4 What type of algorithm does this suggest ?